

T3 group factors: EFT_1_after_F

Active theory: $\text{SM} + S_1 + S_2 + C_{LLS_1S_2}$. The fermion F is absent, so y_1 , y_2 and M_F are not running EFT couplings.

$$j = \frac{d-1}{2}, \quad C_2(d) = \frac{d^2-1}{4}, \quad T(d) = \frac{d(d^2-1)}{12}.$$

In every comparison table, $\mathcal{S}$ denotes the RGE structure multiplying the displayed group factor in $16\pi^2\beta$.

Notation key

symbol	interaction / definition
g_1, g_2, g_3	SM gauge interactions
Y_u, Y_d, Y_e	SM Yukawa interactions
y_1	$LFS_1 + \text{h.c.}$
y_2	$LFS_2 + \text{h.c.}$
M_F	$-\bar{F}M_FF$
T	$\text{Tr}(Y_e^\dagger Y_e + 3Y_u^\dagger Y_u + 3Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\text{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\text{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2, m_2^2	$m_1^2 S_1^\dagger S_1, m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2}\lambda_1(H^\dagger H)^2$
$\lambda_{S_i}^{(1)}$	$\frac{1}{2}\lambda_{S_i}^{(1)}(S_i^\dagger S_i)^2$
$\lambda_{HS_i}^{(1)}$	$(H^\dagger H)(S_i^\dagger S_i)$
$\lambda_{HS_i}^{(A)}$	$(H^\dagger T^A H)(S_i^\dagger T^A S_i)$
λ_5	$HHS_1S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$(S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$(S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	crossed independent S_1 – S_2 contraction

Stage field content

run	S_1	S_2	F	integrated at this stage	active after matching
T3-A	$(1, -1)$	$(3, 0)$	$(2, -\frac{1}{2})$	F	S1, S2
T3-A	$(1, -\frac{1}{2})$	$(3, \frac{1}{2})$	$(2, 0)$	F	S1, S2
T3-A	$(1, 0)$	$(3, 1)$	$(2, \frac{1}{2})$	F	S1, S2
T3-A	$(1, \frac{1}{2})$	$(3, \frac{3}{2})$	$(2, 1)$	F	S1, S2
T3-A	$(1, 1)$	$(3, 2)$	$(2, \frac{3}{2})$	F	S1, S2
T3-B	$(2, -1)$	$(2, 0)$	$(1, -\frac{1}{2})$	F	S1, S2
T3-B	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	$(1, 0)$	F	S1, S2
T3-B	$(2, 0)$	$(2, 1)$	$(1, \frac{1}{2})$	F	S1, S2
T3-B	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	$(1, 1)$	F	S1, S2
T3-B	$(2, 1)$	$(2, 2)$	$(1, \frac{3}{2})$	F	S1, S2
T3-C	$(2, -1)$	$(2, 0)$	$(3, -\frac{1}{2})$	F	S1, S2
T3-C	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	$(3, 0)$	F	S1, S2
T3-C	$(2, 0)$	$(2, 1)$	$(3, \frac{1}{2})$	F	S1, S2
T3-C	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	$(3, 1)$	F	S1, S2
T3-C	$(2, 1)$	$(2, 2)$	$(3, \frac{3}{2})$	F	S1, S2
T3-D	$(3, -1)$	$(1, 0)$	$(2, -\frac{1}{2})$	F	S1, S2
T3-D	$(3, -\frac{1}{2})$	$(1, \frac{1}{2})$	$(2, 0)$	F	S1, S2
T3-D	$(3, 0)$	$(1, 1)$	$(2, \frac{1}{2})$	F	S1, S2
T3-D	$(3, \frac{1}{2})$	$(1, \frac{3}{2})$	$(2, 1)$	F	S1, S2

T3-D	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	F	S1, S2
T3-E	(3, -1)	(3, 0)	(2, $-\frac{1}{2}$)	F	S1, S2
T3-E	(3, $-\frac{1}{2}$)	(3, $\frac{1}{2}$)	(2, 0)	F	S1, S2
T3-E	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	F	S1, S2
T3-E	(3, $\frac{1}{2}$)	(3, $\frac{3}{2}$)	(2, 1)	F	S1, S2
T3-E	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	F	S1, S2

Field representations

run	S_1	S_2	α
T3 - A	(1, -1)	(3, 0)	-2
T3 - A	(1, $-\frac{1}{2}$)	(3, $\frac{1}{2}$)	-1
T3 - A	(1, 0)	(3, 1)	0
T3 - A	(1, $\frac{1}{2}$)	(3, $\frac{3}{2}$)	1
T3 - A	(1, 1)	(3, 2)	2
T3 - B	(2, -1)	(2, 0)	-2
T3 - B	(2, $-\frac{1}{2}$)	(2, $\frac{1}{2}$)	-1
T3 - B	(2, 0)	(2, 1)	0
T3 - B	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	1
T3 - B	(2, 1)	(2, 2)	2
T3 - C	(2, -1)	(2, 0)	-2
T3 - C	(2, $-\frac{1}{2}$)	(2, $\frac{1}{2}$)	-1
T3 - C	(2, 0)	(2, 1)	0
T3 - C	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	1
T3 - C	(2, 1)	(2, 2)	2
T3 - D	(3, -1)	(1, 0)	-2
T3 - D	(3, $-\frac{1}{2}$)	(1, $\frac{1}{2}$)	-1
T3 - D	(3, 0)	(1, 1)	0
T3 - D	(3, $\frac{1}{2}$)	(1, $\frac{3}{2}$)	1
T3 - D	(3, 1)	(1, 2)	2
T3 - E	(3, -1)	(3, 0)	-2
T3 - E	(3, $-\frac{1}{2}$)	(3, $\frac{1}{2}$)	-1
T3 - E	(3, 0)	(3, 1)	0
T3 - E	(3, $\frac{1}{2}$)	(3, $\frac{3}{2}$)	1
T3 - E	(3, 1)	(3, 2)	2

Gauge and Yukawa beta functions in this EFT

$$\beta_{g_1}^{(1)}$$

$\mathcal{S}$	T3 - A ($\alpha = -2$)	T3 - A ($\alpha = -1$)	T3 - A ($\alpha = 0$)	T3 - A ($\alpha = 1$)	T3 - A ($\alpha = 2$)
g_1^3	$\frac{43}{6}$	$\frac{43}{6}$	$\frac{47}{6}$	$\frac{55}{6}$	$\frac{67}{6}$

$\mathcal{S}$	T3 - B ($\alpha = -2$)	T3 - B ($\alpha = -1$)	T3 - B ($\alpha = 0$)	T3 - B ($\alpha = 1$)	T3 - B ($\alpha = 2$)
g_1^3	$\frac{15}{2}$	$\frac{43}{6}$	$\frac{15}{2}$	$\frac{17}{2}$	$\frac{61}{6}$

$\mathcal{S}$	T3 - C ($\alpha = -2$)	T3 - C ($\alpha = -1$)	T3 - C ($\alpha = 0$)	T3 - C ($\alpha = 1$)	T3 - C ($\alpha = 2$)
g_1^3	$\frac{15}{2}$	$\frac{43}{6}$	$\frac{15}{2}$	$\frac{17}{2}$	$\frac{61}{6}$

$\mathcal{S}$	T3 – D ($\alpha = -2$)	T3 – D ($\alpha = -1$)	T3 – D ($\alpha = 0$)	T3 – D ($\alpha = 1$)	T3 – D ($\alpha = 2$)
g_1^3	$\frac{47}{6}$	$\frac{43}{6}$	$\frac{43}{6}$	$\frac{47}{6}$	$\frac{55}{6}$
$\beta_{g_2}^{(1)}$					
$\mathcal{S}$	T3 – E ($\alpha = -2$)	T3 – E ($\alpha = -1$)	T3 – E ($\alpha = 0$)	T3 – E ($\alpha = 1$)	T3 – E ($\alpha = 2$)
g_1^3	$\frac{47}{6}$	$\frac{22}{3}$	$\frac{47}{6}$	$\frac{28}{3}$	$\frac{71}{6}$
$\beta_{g_2}^{(1)}$					
$\mathcal{S}$	T3 – A ($\alpha = -2$)	T3 – A ($\alpha = -1$)	T3 – A ($\alpha = 0$)	T3 – A ($\alpha = 1$)	T3 – A ($\alpha = 2$)
g_2^3	$-\frac{5}{2}$	$-\frac{5}{2}$	$-\frac{5}{2}$	$-\frac{5}{2}$	$-\frac{5}{2}$
$\beta_{g_2}^{(1)}$					
$\mathcal{S}$	T3 – B ($\alpha = -2$)	T3 – B ($\alpha = -1$)	T3 – B ($\alpha = 0$)	T3 – B ($\alpha = 1$)	T3 – B ($\alpha = 2$)
g_2^3	$-\frac{17}{6}$	$-\frac{17}{6}$	$-\frac{17}{6}$	$-\frac{17}{6}$	$-\frac{17}{6}$
$\beta_{g_2}^{(1)}$					
$\mathcal{S}$	T3 – C ($\alpha = -2$)	T3 – C ($\alpha = -1$)	T3 – C ($\alpha = 0$)	T3 – C ($\alpha = 1$)	T3 – C ($\alpha = 2$)
g_2^3	$-\frac{17}{6}$	$-\frac{17}{6}$	$-\frac{17}{6}$	$-\frac{17}{6}$	$-\frac{17}{6}$
$\beta_{g_2}^{(1)}$					
$\mathcal{S}$	T3 – D ($\alpha = -2$)	T3 – D ($\alpha = -1$)	T3 – D ($\alpha = 0$)	T3 – D ($\alpha = 1$)	T3 – D ($\alpha = 2$)
g_2^3	$-\frac{5}{2}$	$-\frac{5}{2}$	$-\frac{5}{2}$	$-\frac{5}{2}$	$-\frac{5}{2}$
$\beta_{g_2}^{(1)}$					
$\mathcal{S}$	T3 – E ($\alpha = -2$)	T3 – E ($\alpha = -1$)	T3 – E ($\alpha = 0$)	T3 – E ($\alpha = 1$)	T3 – E ($\alpha = 2$)
g_2^3	$-\frac{11}{6}$	$-\frac{11}{6}$	$-\frac{11}{6}$	$-\frac{11}{6}$	$-\frac{11}{6}$
$\beta_{g_3}^{(1)}$					
$\mathcal{S}$	T3 – A ($\alpha = -2$)	T3 – A ($\alpha = -1$)	T3 – A ($\alpha = 0$)	T3 – A ($\alpha = 1$)	T3 – A ($\alpha = 2$)
g_3^3	-7	-7	-7	-7	-7
$\beta_{g_3}^{(1)}$					
$\mathcal{S}$	T3 – B ($\alpha = -2$)	T3 – B ($\alpha = -1$)	T3 – B ($\alpha = 0$)	T3 – B ($\alpha = 1$)	T3 – B ($\alpha = 2$)
g_3^3	-7	-7	-7	-7	-7
$\beta_{g_3}^{(1)}$					
$\mathcal{S}$	T3 – C ($\alpha = -2$)	T3 – C ($\alpha = -1$)	T3 – C ($\alpha = 0$)	T3 – C ($\alpha = 1$)	T3 – C ($\alpha = 2$)
g_3^3	-7	-7	-7	-7	-7
$\beta_{g_3}^{(1)}$					
$\mathcal{S}$	T3 – D ($\alpha = -2$)	T3 – D ($\alpha = -1$)	T3 – D ($\alpha = 0$)	T3 – D ($\alpha = 1$)	T3 – D ($\alpha = 2$)
g_3^3	-7	-7	-7	-7	-7
$\beta_{g_3}^{(1)}$					
$\mathcal{S}$	T3 – E ($\alpha = -2$)	T3 – E ($\alpha = -1$)	T3 – E ($\alpha = 0$)	T3 – E ($\alpha = 1$)	T3 – E ($\alpha = 2$)

g_3^3	-7	-7	-7	-7	-7
---------	----	----	----	----	----

$\beta_{Y_u}^{(1)}$

$\mathcal{S}$	T3 - A ($\alpha = -2$)	T3 - A ($\alpha = -1$)	T3 - A ($\alpha = 0$)	T3 - A ($\alpha = 1$)	T3 - A ($\alpha = 2$)
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_u)_{ij}$	-8	-8	-8	-8	-8
$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$
$T(Y_u)_{ij}$	1	1	1	1	1

$\mathcal{S}$	T3 - B ($\alpha = -2$)	T3 - B ($\alpha = -1$)	T3 - B ($\alpha = 0$)	T3 - B ($\alpha = 1$)	T3 - B ($\alpha = 2$)
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_u)_{ij}$	-8	-8	-8	-8	-8
$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$
$T(Y_u)_{ij}$	1	1	1	1	1

$\mathcal{S}$	T3 - C ($\alpha = -2$)	T3 - C ($\alpha = -1$)	T3 - C ($\alpha = 0$)	T3 - C ($\alpha = 1$)	T3 - C ($\alpha = 2$)
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_u)_{ij}$	-8	-8	-8	-8	-8
$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$
$T(Y_u)_{ij}$	1	1	1	1	1

$\mathcal{S}$	T3 - D ($\alpha = -2$)	T3 - D ($\alpha = -1$)	T3 - D ($\alpha = 0$)	T3 - D ($\alpha = 1$)	T3 - D ($\alpha = 2$)
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_u)_{ij}$	-8	-8	-8	-8	-8
$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$
$T(Y_u)_{ij}$	1	1	1	1	1

$\mathcal{S}$	T3 - E ($\alpha = -2$)	T3 - E ($\alpha = -1$)	T3 - E ($\alpha = 0$)	T3 - E ($\alpha = 1$)	T3 - E ($\alpha = 2$)
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_u)_{ij}$	-8	-8	-8	-8	-8
$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$
$T(Y_u)_{ij}$	1	1	1	1	1

$\beta_{Y_d}^{(1)}$

$\mathcal{S}$	T3 – B ($\alpha = -2$)	T3 – B ($\alpha = -1$)	T3 – B ($\alpha = 0$)	T3 – B ($\alpha = 1$)	T3 – B ($\alpha = 2$)
$(Y_e(Y_e)^\dagger Y_e)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_e)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_1^2(Y_e)_{ij}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$
$T(Y_e)_{ij}$	1	1	1	1	1

$\mathcal{S}$	T3 – C ($\alpha = -2$)	T3 – C ($\alpha = -1$)	T3 – C ($\alpha = 0$)	T3 – C ($\alpha = 1$)	T3 – C ($\alpha = 2$)
$(Y_e(Y_e)^\dagger Y_e)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_e)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_1^2(Y_e)_{ij}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$
$T(Y_e)_{ij}$	1	1	1	1	1

$\mathcal{S}$	T3 – D ($\alpha = -2$)	T3 – D ($\alpha = -1$)	T3 – D ($\alpha = 0$)	T3 – D ($\alpha = 1$)	T3 – D ($\alpha = 2$)
$(Y_e(Y_e)^\dagger Y_e)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_e)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_1^2(Y_e)_{ij}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$
$T(Y_e)_{ij}$	1	1	1	1	1

$\mathcal{S}$	T3 – E ($\alpha = -2$)	T3 – E ($\alpha = -1$)	T3 – E ($\alpha = 0$)	T3 – E ($\alpha = 1$)	T3 – E ($\alpha = 2$)
$(Y_e(Y_e)^\dagger Y_e)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_e)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_1^2(Y_e)_{ij}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$
$T(Y_e)_{ij}$	1	1	1	1	1

$\beta_{m_1^2}$

Run block 1: T3-A–T3-A

$\mathcal{S}$	T3 – A ($\alpha = -2$)	T3 – A ($\alpha = -1$)	T3 – A ($\alpha = 0$)	T3 – A ($\alpha = 1$)	T3 – A ($\alpha = 2$)
$\lambda_{S_1}^{(1)} m_1^2$	4	4	4	4	4
$g_2^2 m_1^2$	0	0	0	0	0
$g_1^2 m_1^2$	-6	$-\frac{3}{2}$	0	$-\frac{3}{2}$	-6
$\lambda_{12}^{(1)} m_2^2$	6	6	6	6	6
$\lambda_{S_1}^{(A)} m_1^2$	0	0	0	0	0
$\lambda_{12}^{(\times)} m_2^2$	0	0	0	0	0

Run block 2: T3-B–T3-B

$\mathcal{S}$	T3 – B ($\alpha = -2$)	T3 – B ($\alpha = -1$)	T3 – B ($\alpha = 0$)	T3 – B ($\alpha = 1$)	T3 – B ($\alpha = 2$)
$\lambda_{S_1}^{(1)} m_1^2$	6	6	6	6	6
$g_2^2 m_1^2$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$g_1^2 m_1^2$	-6	$-\frac{3}{2}$	0	$-\frac{3}{2}$	-6
$\lambda_{12}^{(1)} m_2^2$	4	4	4	4	4
$\lambda_{S_1}^{(A)} m_1^2$	0	0	0	0	0
$\lambda_{12}^{(\times)} m_2^2$	0	0	0	0	0

Run block 3: T3-C–T3-C

$\mathcal{S}$	T3 – C ($\alpha = -2$)	T3 – C ($\alpha = -1$)	T3 – C ($\alpha = 0$)	T3 – C ($\alpha = 1$)	T3 – C ($\alpha = 2$)
$\lambda_{S_1}^{(1)} m_1^2$	6	6	6	6	6
$g_2^2 m_1^2$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$g_1^2 m_1^2$	-6	$-\frac{3}{2}$	0	$-\frac{3}{2}$	-6
$\lambda_{12}^{(1)} m_2^2$	4	4	4	4	4
$\lambda_{S_1}^{(A)} m_1^2$	0	0	0	0	0
$\lambda_{12}^{(\times)} m_2^2$	0	0	0	0	0

Run block 4: T3-D–T3-D

$\mathcal{S}$	T3 – D ($\alpha = -2$)	T3 – D ($\alpha = -1$)	T3 – D ($\alpha = 0$)	T3 – D ($\alpha = 1$)	T3 – D ($\alpha = 2$)
$\lambda_{S_1}^{(1)} m_1^2$	8	8	8	8	8
$g_2^2 m_1^2$	-12	-12	-12	-12	-12
$g_1^2 m_1^2$	-6	$-\frac{3}{2}$	0	$-\frac{3}{2}$	-6
$\lambda_{12}^{(1)} m_2^2$	2	2	2	2	2
$\lambda_{S_1}^{(A)} m_1^2$	4	4	4	4	4
$\lambda_{12}^{(\times)} m_2^2$	0	0	0	0	0

Run block 5: T3-E–T3-E

$\mathcal{S}$	T3 – E ($\alpha = -2$)	T3 – E ($\alpha = -1$)	T3 – E ($\alpha = 0$)	T3 – E ($\alpha = 1$)	T3 – E ($\alpha = 2$)
$\lambda_{S_1}^{(1)} m_1^2$	8	8	8	8	8
$g_2^2 m_1^2$	-12	-12	-12	-12	-12
$g_1^2 m_1^2$	-6	$-\frac{3}{2}$	0	$-\frac{3}{2}$	-6
$\lambda_{12}^{(1)} m_2^2$	6	6	6	6	6
$\lambda_{S_1}^{(A)} m_1^2$	4	4	4	4	4
$\lambda_{12}^{(\times)} m_2^2$	4	4	4	4	4

$\beta_{m_2^2}$

Run block 1: T3-A–T3-A

$\mathcal{S}$	T3 – A ($\alpha = -2$)	T3 – A ($\alpha = -1$)	T3 – A ($\alpha = 0$)	T3 – A ($\alpha = 1$)	T3 – A ($\alpha = 2$)
$\lambda_{S_2}^{(1)} m_2^2$	8	8	8	8	8
$g_2^2 m_2^2$	-12	-12	-12	-12	-12
$g_1^2 m_2^2$	0	$-\frac{3}{2}$	-6	$-\frac{27}{2}$	-24
$\lambda_{12}^{(1)} m_1^2$	2	2	2	2	2
$\lambda_{S_2}^{(A)} m_2^2$	4	4	4	4	4
$\lambda_{12}^{(\times)} m_1^2$	0	0	0	0	0

Run block 2: T3-B–T3-B

$\mathcal{S}$	T3 – B ($\alpha = -2$)	T3 – B ($\alpha = -1$)	T3 – B ($\alpha = 0$)	T3 – B ($\alpha = 1$)	T3 – B ($\alpha = 2$)
$\lambda_{S_2}^{(1)} m_2^2$	6	6	6	6	6
$g_2^2 m_2^2$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$g_1^2 m_2^2$	0	$-\frac{3}{2}$	-6	$-\frac{27}{2}$	-24
$\lambda_{12}^{(1)} m_1^2$	4	4	4	4	4
$\lambda_{S_2}^{(A)} m_2^2$	0	0	0	0	0
$\lambda_{12}^{(\times)} m_1^2$	0	0	0	0	0

Run block 3: T3-C–T3-C

$\mathcal{S}$	T3 – C ($\alpha = -2$)	T3 – C ($\alpha = -1$)	T3 – C ($\alpha = 0$)	T3 – C ($\alpha = 1$)	T3 – C ($\alpha = 2$)
$\lambda_{S_2}^{(1)} m_2^2$	6	6	6	6	6
$g_2^2 m_2^2$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$g_1^2 m_2^2$	0	$-\frac{3}{2}$	-6	$-\frac{27}{2}$	-24
$\lambda_{12}^{(1)} m_1^2$	4	4	4	4	4
$\lambda_{S_2}^{(A)} m_2^2$	0	0	0	0	0
$\lambda_{12}^{(\times)} m_1^2$	0	0	0	0	0

Run block 4: T3-D–T3-D

$\mathcal{S}$	T3 – D ($\alpha = -2$)	T3 – D ($\alpha = -1$)	T3 – D ($\alpha = 0$)	T3 – D ($\alpha = 1$)	T3 – D ($\alpha = 2$)
$\lambda_{S_2}^{(1)} m_2^2$	4	4	4	4	4
$g_2^2 m_2^2$	0	0	0	0	0
$g_1^2 m_2^2$	0	$-\frac{3}{2}$	-6	$-\frac{27}{2}$	-24
$\lambda_{12}^{(1)} m_1^2$	6	6	6	6	6
$\lambda_{S_2}^{(A)} m_2^2$	0	0	0	0	0
$\lambda_{12}^{(\times)} m_1^2$	0	0	0	0	0

Run block 5: T3-E–T3-E

$\mathcal{S}$	T3 – E ($\alpha = -2$)	T3 – E ($\alpha = -1$)	T3 – E ($\alpha = 0$)	T3 – E ($\alpha = 1$)	T3 – E ($\alpha = 2$)
$\lambda_{S_2}^{(1)} m_2^2$	8	8	8	8	8
$g_2^2 m_2^2$	-12	-12	-12	-12	-12
$g_1^2 m_2^2$	0	$-\frac{3}{2}$	-6	$-\frac{27}{2}$	-24
$\lambda_{12}^{(1)} m_1^2$	6	6	6	6	6
$\lambda_{S_2}^{(A)} m_2^2$	4	4	4	4	4
$\lambda_{12}^{(\times)} m_1^2$	4	4	4	4	4

$\beta_{\lambda_{HS_1}^{(1)}}$

Run block 1: T3-A–T3-A

$\mathcal{S}$	T3 – A ($\alpha = -2$)	T3 – A ($\alpha = -1$)	T3 – A ($\alpha = 0$)	T3 – A ($\alpha = 1$)	T3 – A ($\alpha = 2$)
$ \lambda_5 ^2$	12	12	12	12	12
$g_2^2 \lambda_{HS_1}^{(1)}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$g_1^2 \lambda_{HS_1}^{(1)}$	$-\frac{15}{2}$	-3	$-\frac{3}{2}$	-3	$-\frac{15}{2}$
$\lambda_1 \lambda_{HS_1}^{(1)}$	6	6	6	6	6
$\left(\lambda_{HS_1}^{(1)}\right)^2$	4	4	4	4	4
$\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(1)}$	4	4	4	4	4
$\lambda_{12}^{(1)} \lambda_{HS_2}^{(1)}$	6	6	6	6	6
g_2^4	0	0	0	0	0
g_1^4	3	$\frac{3}{4}$	0	$\frac{3}{4}$	3
$\lambda_{HS_1}^{(1)} T$	2	2	2	2	2
$\left(\lambda_{HS_1}^{(A)}\right)^2$	0	0	0	0	0
$\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(A)}$	0	0	0	0	0
$\lambda_{HS_2}^{(1)} \lambda_{12}^{(\times)}$	0	0	0	0	0

Run block 2: T3-B–T3-B

$\mathcal{S}$	T3 – B ($\alpha = -2$)	T3 – B ($\alpha = -1$)	T3 – B ($\alpha = 0$)	T3 – B ($\alpha = 1$)	T3 – B ($\alpha = 2$)
$ \lambda_5 ^2$	6	6	6	6	6
$g_2^2 \lambda_{HS_1}^{(1)}$	-9	-9	-9	-9	-9
$g_1^2 \lambda_{HS_1}^{(1)}$	$-\frac{15}{2}$	-3	$-\frac{3}{2}$	-3	$-\frac{15}{2}$
$\lambda_1 \lambda_{HS_1}^{(1)}$	6	6	6	6	6
$\left(\lambda_{HS_1}^{(1)}\right)^2$	4	4	4	4	4
$\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(1)}$	6	6	6	6	6
$\lambda_{12}^{(1)} \lambda_{HS_2}^{(1)}$	4	4	4	4	4
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
g_1^4	3	$\frac{3}{4}$	0	$\frac{3}{4}$	3
$\lambda_{HS_1}^{(1)} T$	2	2	2	2	2
$\left(\lambda_{HS_1}^{(A)}\right)^2$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(A)}$	0	0	0	0	0
$\lambda_{HS_2}^{(1)} \lambda_{12}^{(\times)}$	0	0	0	0	0

Run block 3: T3-C-T3-C

$\mathcal{S}$	T3 – C ($\alpha = -2$)	T3 – C ($\alpha = -1$)	T3 – C ($\alpha = 0$)	T3 – C ($\alpha = 1$)	T3 – C ($\alpha = 2$)
$ \lambda_5 ^2$	6	6	6	6	6
$g_2^2 \lambda_{HS_1}^{(1)}$	-9	-9	-9	-9	-9
$g_1^2 \lambda_{HS_1}^{(1)}$	$-\frac{15}{2}$	-3	$-\frac{3}{2}$	-3	$-\frac{15}{2}$
$\lambda_1 \lambda_{HS_1}^{(1)}$	6	6	6	6	6
$\left(\lambda_{HS_1}^{(1)}\right)^2$	4	4	4	4	4
$\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(1)}$	6	6	6	6	6
$\lambda_{12}^{(1)} \lambda_{HS_2}^{(1)}$	4	4	4	4	4
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
g_1^4	3	$\frac{3}{4}$	0	$\frac{3}{4}$	3
$\lambda_{HS_1}^{(1)} T$	2	2	2	2	2
$\left(\lambda_{HS_1}^{(A)}\right)^2$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(A)}$	0	0	0	0	0
$\lambda_{HS_2}^{(1)} \lambda_{12}^{(\times)}$	0	0	0	0	0

Run block 4: T3-D-T3-D

$\mathcal{S}$	T3 – D ($\alpha = -2$)	T3 – D ($\alpha = -1$)	T3 – D ($\alpha = 0$)	T3 – D ($\alpha = 1$)	T3 – D ($\alpha = 2$)
$ \lambda_5 ^2$	4	4	4	4	4
$g_2^2 \lambda_{HS_1}^{(1)}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$
$g_1^2 \lambda_{HS_1}^{(1)}$	$-\frac{15}{2}$	-3	$-\frac{3}{2}$	-3	$-\frac{15}{2}$
$\lambda_1 \lambda_{HS_1}^{(1)}$	6	6	6	6	6
$\left(\lambda_{HS_1}^{(1)}\right)^2$	4	4	4	4	4
$\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(1)}$	8	8	8	8	8
$\lambda_{12}^{(1)} \lambda_{HS_2}^{(1)}$	2	2	2	2	2
g_2^4	6	6	6	6	6
g_1^4	3	$\frac{3}{4}$	0	$\frac{3}{4}$	3
$\lambda_{HS_1}^{(1)} T$	2	2	2	2	2
$\left(\lambda_{HS_1}^{(A)}\right)^2$	2	2	2	2	2
$\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(A)}$	4	4	4	4	4
$\lambda_{HS_2}^{(1)} \lambda_{12}^{(\times)}$	0	0	0	0	0

Run block 5: T3-E-T3-E

$\mathcal{S}$	T3 – E ($\alpha = -2$)	T3 – E ($\alpha = -1$)	T3 – E ($\alpha = 0$)	T3 – E ($\alpha = 1$)	T3 – E ($\alpha = 2$)
$ \lambda_5 ^2$	4	4	4	4	4
$g_2^2 \lambda_{HS_1}^{(1)}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$
$g_1^2 \lambda_{HS_1}^{(1)}$	$-\frac{15}{2}$	-3	$-\frac{3}{2}$	-3	$-\frac{15}{2}$
$\lambda_1 \lambda_{HS_1}^{(1)}$	6	6	6	6	6
$\left(\lambda_{HS_1}^{(1)}\right)^2$	4	4	4	4	4
$\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(1)}$	8	8	8	8	8
$\lambda_{12}^{(1)} \lambda_{HS_2}^{(1)}$	6	6	6	6	6
g_2^4	6	6	6	6	6
g_1^4	3	$\frac{3}{4}$	0	$\frac{3}{4}$	3
$\lambda_{HS_1}^{(1)} T$	2	2	2	2	2
$\left(\lambda_{HS_1}^{(A)}\right)^2$	2	2	2	2	2
$\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(A)}$	4	4	4	4	4
$\lambda_{HS_2}^{(1)} \lambda_{12}^{(\times)}$	4	4	4	4	4

$\beta_{\lambda_{HS_2}^{(1)}}$

Run block 1: T3-A-T3-A

$\mathcal{S}$	T3 – A ($\alpha = -2$)	T3 – A ($\alpha = -1$)	T3 – A ($\alpha = 0$)	T3 – A ($\alpha = 1$)	T3 – A ($\alpha = 2$)
$ \lambda_5 ^2$	4	4	4	4	4
$g_2^2 \lambda_{HS_2}^{(1)}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$
$g_1^2 \lambda_{HS_2}^{(1)}$	$-\frac{3}{2}$	-3	$-\frac{15}{2}$	-15	$-\frac{51}{2}$
$\lambda_1 \lambda_{HS_2}^{(1)}$	6	6	6	6	6
$\left(\lambda_{HS_2}^{(1)}\right)^2$	4	4	4	4	4
$\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(1)}$	8	8	8	8	8
$\lambda_{12}^{(1)} \lambda_{HS_1}^{(1)}$	2	2	2	2	2
g_2^4	6	6	6	6	6
g_1^4	0	$\frac{3}{4}$	3	$\frac{27}{4}$	12
$\left(\lambda_{HS_2}^{(A)}\right)^2$	2	2	2	2	2
$\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(A)}$	4	4	4	4	4
$\lambda_{HS_2}^{(1)} T$	2	2	2	2	2
$\lambda_{HS_1}^{(1)} \lambda_{12}^{(\times)}$	0	0	0	0	0

Run block 2: T3-B-T3-B

$\mathcal{S}$	T3 – B ($\alpha = -2$)	T3 – B ($\alpha = -1$)	T3 – B ($\alpha = 0$)	T3 – B ($\alpha = 1$)	T3 – B ($\alpha = 2$)
$ \lambda_5 ^2$	6	6	6	6	6
$g_2^2 \lambda_{HS_2}^{(1)}$	-9	-9	-9	-9	-9
$g_1^2 \lambda_{HS_2}^{(1)}$	$-\frac{3}{2}$	-3	$-\frac{15}{2}$	-15	$-\frac{51}{2}$
$\lambda_1 \lambda_{HS_2}^{(1)}$	6	6	6	6	6
$\left(\lambda_{HS_2}^{(1)}\right)^2$	4	4	4	4	4
$\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(1)}$	6	6	6	6	6
$\lambda_{12}^{(1)} \lambda_{HS_1}^{(1)}$	4	4	4	4	4
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
g_1^4	0	$\frac{3}{4}$	3	$\frac{27}{4}$	12
$\left(\lambda_{HS_2}^{(A)}\right)^2$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(A)}$	0	0	0	0	0
$\lambda_{HS_2}^{(1)} T$	2	2	2	2	2

$\lambda_{HS_1}^{(1)} \lambda_{12}^{(\times)}$	0	0	0	0	0
--	---	---	---	---	---

Run block 3: T3-C-T3-C

$\mathcal{S}$	T3 – C ($\alpha = -2$)	T3 – C ($\alpha = -1$)	T3 – C ($\alpha = 0$)	T3 – C ($\alpha = 1$)	T3 – C ($\alpha = 2$)
$ \lambda_5 ^2$	6	6	6	6	6
$g_2^2 \lambda_{HS_2}^{(1)}$	-9	-9	-9	-9	-9
$g_1^2 \lambda_{HS_2}^{(1)}$	$-\frac{3}{2}$	-3	$-\frac{15}{2}$	-15	$-\frac{51}{2}$
$\lambda_1 \lambda_{HS_2}^{(1)}$	6	6	6	6	6
$\left(\lambda_{HS_2}^{(1)}\right)^2$	4	4	4	4	4
$\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(1)}$	6	6	6	6	6
$\lambda_{12}^{(1)} \lambda_{HS_1}^{(1)}$	4	4	4	4	4
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
g_1^4	0	$\frac{3}{4}$	3	$\frac{27}{4}$	12
$\left(\lambda_{HS_2}^{(A)}\right)^2$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(A)}$	0	0	0	0	0
$\lambda_{HS_2}^{(1)} T$	2	2	2	2	2
$\lambda_{HS_1}^{(1)} \lambda_{12}^{(\times)}$	0	0	0	0	0

Run block 4: T3-D-T3-D

$\mathcal{S}$	T3 – D ($\alpha = -2$)	T3 – D ($\alpha = -1$)	T3 – D ($\alpha = 0$)	T3 – D ($\alpha = 1$)	T3 – D ($\alpha = 2$)
$ \lambda_5 ^2$	12	12	12	12	12
$g_2^2 \lambda_{HS_2}^{(1)}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$g_1^2 \lambda_{HS_2}^{(1)}$	$-\frac{3}{2}$	-3	$-\frac{15}{2}$	-15	$-\frac{51}{2}$
$\lambda_1 \lambda_{HS_2}^{(1)}$	6	6	6	6	6
$\left(\lambda_{HS_2}^{(1)}\right)^2$	4	4	4	4	4
$\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(1)}$	4	4	4	4	4
$\lambda_{12}^{(1)} \lambda_{HS_1}^{(1)}$	6	6	6	6	6
g_2^4	0	0	0	0	0
g_1^4	0	$\frac{3}{4}$	3	$\frac{27}{4}$	12
$\left(\lambda_{HS_2}^{(A)}\right)^2$	0	0	0	0	0
$\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(A)}$	0	0	0	0	0
$\lambda_{HS_2}^{(1)} T$	2	2	2	2	2
$\lambda_{HS_1}^{(1)} \lambda_{12}^{(\times)}$	0	0	0	0	0

Run block 5: T3-E-T3-E

$\mathcal{S}$	T3 – E ($\alpha = -2$)	T3 – E ($\alpha = -1$)	T3 – E ($\alpha = 0$)	T3 – E ($\alpha = 1$)	T3 – E ($\alpha = 2$)
$ \lambda_5 ^2$	4	4	4	4	4
$g_2^2 \lambda_{HS_2}^{(1)}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$
$g_1^2 \lambda_{HS_2}^{(1)}$	$-\frac{3}{2}$	-3	$-\frac{15}{2}$	-15	$-\frac{51}{2}$
$\lambda_1 \lambda_{HS_2}^{(1)}$	6	6	6	6	6
$\left(\lambda_{HS_2}^{(1)}\right)^2$	4	4	4	4	4
$\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(1)}$	8	8	8	8	8
$\lambda_{12}^{(1)} \lambda_{HS_1}^{(1)}$	6	6	6	6	6
g_2^4	6	6	6	6	6
g_1^4	0	$\frac{3}{4}$	3	$\frac{27}{4}$	12
$\left(\lambda_{HS_2}^{(A)}\right)^2$	2	2	2	2	2
$\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(A)}$	4	4	4	4	4

$\lambda_{HS_2}^{(1)} T$	2	2	2	2	2
$\lambda_{HS_1}^{(1)} \lambda_{12}^{(\times)}$	4	4	4	4	4

$$\beta_{\lambda_{12}^{(1)}}$$

Run block 1: T3-A-T3-A

$\mathcal{S}$	T3 – A ($\alpha = -2$)	T3 – A ($\alpha = -1$)	T3 – A ($\alpha = 0$)	T3 – A ($\alpha = 1$)	T3 – A ($\alpha = 2$)
$ \lambda_5 ^2$	4	4	4	4	4
$g_2^2 \lambda_{12}^{(1)}$	-12	-12	-12	-12	-12
$g_1^2 \lambda_{12}^{(1)}$	-6	-3	-6	-15	-30
$\left(\lambda_{12}^{(1)}\right)^2$	4	4	4	4	4
$\lambda_{12}^{(1)} \lambda_{S_1}^{(1)}$	4	4	4	4	4
$\lambda_{12}^{(1)} \lambda_{S_2}^{(1)}$	8	8	8	8	8
$\lambda_{HS_1}^{(1)} \lambda_{HS_2}^{(1)}$	4	4	4	4	4
g_2^4	0	0	0	0	0
g_1^4	0	$\frac{3}{4}$	0	$\frac{27}{4}$	48
$\lambda_{12}^{(1)} \lambda_{S_2}^{(A)}$	4	4	4	4	4
$\left(\lambda_{12}^{(A)}\right)^2$	0	0	0	0	0
$\lambda_{12}^{(1)} \lambda_{S_1}^{(A)}$	0	0	0	0	0
$\left(\lambda_{12}^{(\times)}\right)^2$	0	0	0	0	0
$\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$	0	0	0	0	0
$\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$	0	0	0	0	0

Run block 2: T3-B-T3-B

$\mathcal{S}$	T3 – B ($\alpha = -2$)	T3 – B ($\alpha = -1$)	T3 – B ($\alpha = 0$)	T3 – B ($\alpha = 1$)	T3 – B ($\alpha = 2$)
$ \lambda_5 ^2$	3	3	3	3	3
$g_2^2 \lambda_{12}^{(1)}$	-9	-9	-9	-9	-9
$g_1^2 \lambda_{12}^{(1)}$	-6	-3	-6	-15	-30
$\left(\lambda_{12}^{(1)}\right)^2$	4	4	4	4	4
$\lambda_{12}^{(1)} \lambda_{S_1}^{(1)}$	6	6	6	6	6
$\lambda_{12}^{(1)} \lambda_{S_2}^{(1)}$	6	6	6	6	6
$\lambda_{HS_1}^{(1)} \lambda_{HS_2}^{(1)}$	4	4	4	4	4
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
g_1^4	0	$\frac{3}{4}$	0	$\frac{27}{4}$	48
$\lambda_{12}^{(1)} \lambda_{S_2}^{(A)}$	0	0	0	0	0
$\left(\lambda_{12}^{(A)}\right)^2$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$\lambda_{12}^{(1)} \lambda_{S_1}^{(A)}$	0	0	0	0	0
$\left(\lambda_{12}^{(\times)}\right)^2$	0	0	0	0	0
$\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$	0	0	0	0	0
$\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$	0	0	0	0	0

Run block 3: T3-C-T3-C

$\mathcal{S}$	T3 – C ($\alpha = -2$)	T3 – C ($\alpha = -1$)	T3 – C ($\alpha = 0$)	T3 – C ($\alpha = 1$)	T3 – C ($\alpha = 2$)
$ \lambda_5 ^2$	3	3	3	3	3
$g_2^2 \lambda_{12}^{(1)}$	-9	-9	-9	-9	-9
$g_1^2 \lambda_{12}^{(1)}$	-6	-3	-6	-15	-30

$\left(\lambda_{12}^{(1)}\right)^2$	4	4	4	4	4
$\lambda_{12}^{(1)} \lambda_{S_1}^{(1)}$	6	6	6	6	6
$\lambda_{12}^{(1)} \lambda_{S_2}^{(1)}$	6	6	6	6	6
$\lambda_{HS_1}^{(1)} \lambda_{HS_2}^{(1)}$	4	4	4	4	4
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
g_1^4	0	$\frac{3}{4}$	0	$\frac{27}{4}$	48
$\lambda_{12}^{(1)} \lambda_{S_2}^{(A)}$	0	0	0	0	0
$\left(\lambda_{12}^{(A)}\right)^2$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$\lambda_{12}^{(1)} \lambda_{S_1}^{(A)}$	0	0	0	0	0
$\left(\lambda_{12}^{(\times)}\right)^2$	0	0	0	0	0
$\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$	0	0	0	0	0
$\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$	0	0	0	0	0

Run block 4: T3-D-T3-D

$\mathcal{S}$	T3 – D ($\alpha = -2$)	T3 – D ($\alpha = -1$)	T3 – D ($\alpha = 0$)	T3 – D ($\alpha = 1$)	T3 – D ($\alpha = 2$)
$ \lambda_5 ^2$	4	4	4	4	4
$g_2^2 \lambda_{12}^{(1)}$	-12	-12	-12	-12	-12
$g_1^2 \lambda_{12}^{(1)}$	-6	-3	-6	-15	-30
$\left(\lambda_{12}^{(1)}\right)^2$	4	4	4	4	4
$\lambda_{12}^{(1)} \lambda_{S_1}^{(1)}$	8	8	8	8	8
$\lambda_{12}^{(1)} \lambda_{S_2}^{(1)}$	4	4	4	4	4
$\lambda_{HS_1}^{(1)} \lambda_{HS_2}^{(1)}$	4	4	4	4	4
g_2^4	0	0	0	0	0
g_1^4	0	$\frac{3}{4}$	0	$\frac{27}{4}$	48
$\lambda_{12}^{(1)} \lambda_{S_2}^{(A)}$	0	0	0	0	0
$\left(\lambda_{12}^{(A)}\right)^2$	0	0	0	0	0
$\lambda_{12}^{(1)} \lambda_{S_1}^{(A)}$	4	4	4	4	4
$\left(\lambda_{12}^{(\times)}\right)^2$	0	0	0	0	0
$\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$	0	0	0	0	0
$\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$	0	0	0	0	0

Run block 5: T3-E-T3-E

$\mathcal{S}$	T3 – E ($\alpha = -2$)	T3 – E ($\alpha = -1$)	T3 – E ($\alpha = 0$)	T3 – E ($\alpha = 1$)	T3 – E ($\alpha = 2$)
$ \lambda_5 ^2$	$\frac{4}{3}$	$\frac{4}{3}$	$\frac{4}{3}$	$\frac{4}{3}$	$\frac{4}{3}$
$g_2^2 \lambda_{12}^{(1)}$	-24	-24	-24	-24	-24
$g_1^2 \lambda_{12}^{(1)}$	-6	-3	-6	-15	-30
$\left(\lambda_{12}^{(1)}\right)^2$	4	4	4	4	4
$\lambda_{12}^{(1)} \lambda_{S_1}^{(1)}$	8	8	8	8	8
$\lambda_{12}^{(1)} \lambda_{S_2}^{(1)}$	8	8	8	8	8
$\lambda_{HS_1}^{(1)} \lambda_{HS_2}^{(1)}$	4	4	4	4	4
g_2^4	12	12	12	12	12
g_1^4	0	$\frac{3}{4}$	0	$\frac{27}{4}$	48
$\lambda_{12}^{(1)} \lambda_{S_2}^{(A)}$	4	4	4	4	4
$\left(\lambda_{12}^{(A)}\right)^2$	4	4	4	4	4
$\lambda_{12}^{(1)} \lambda_{S_1}^{(A)}$	4	4	4	4	4
$\left(\lambda_{12}^{(\times)}\right)^2$	4	4	4	4	4
$\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$	4	4	4	4	4

$\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$	4	4	4	4	4
---	---	---	---	---	---

β_{λ_5}

Run block 1: T3-A-T3-A

$\mathcal{S}$	T3 - A ($\alpha = -2$)	T3 - A ($\alpha = -1$)	T3 - A ($\alpha = 0$)	T3 - A ($\alpha = 1$)	T3 - A ($\alpha = 2$)
$\lambda_5 \lambda_1$	2	2	2	2	2
$\lambda_5 \lambda_{HS_1}^{(1)}$	4	4	4	4	4
$\lambda_5 \lambda_{HS_2}^{(1)}$	4	4	4	4	4
$\lambda_5 \lambda_{12}^{(1)}$	2	2	2	2	2
$g_2^2 \lambda_5$	$-\frac{21}{2}$	$-\frac{21}{2}$	$-\frac{21}{2}$	$-\frac{21}{2}$	$-\frac{21}{2}$
$g_1^2 \lambda_5$	$-\frac{9}{2}$	-3	$-\frac{9}{2}$	-9	$-\frac{33}{2}$
$\lambda_5 \lambda_{HS_2}^{(A)}$	4	4	4	4	4
$\lambda_5 T$	2	2	2	2	2
$\lambda_5 \lambda_{HS_1}^{(A)}$	0	0	0	0	0
$\lambda_5 \lambda_{12}^{(A)}$	0	0	0	0	0
$\lambda_5 \lambda_{12}^{(\times)}$	0	0	0	0	0

Run block 2: T3-B-T3-B

$\mathcal{S}$	T3 - B ($\alpha = -2$)	T3 - B ($\alpha = -1$)	T3 - B ($\alpha = 0$)	T3 - B ($\alpha = 1$)	T3 - B ($\alpha = 2$)
$\lambda_5 \lambda_1$	2	2	2	2	2
$\lambda_5 \lambda_{HS_1}^{(1)}$	4	4	4	4	4
$\lambda_5 \lambda_{HS_2}^{(1)}$	4	4	4	4	4
$\lambda_5 \lambda_{12}^{(1)}$	2	2	2	2	2
$g_2^2 \lambda_5$	-9	-9	-9	-9	-9
$g_1^2 \lambda_5$	$-\frac{9}{2}$	-3	$-\frac{9}{2}$	-9	$-\frac{33}{2}$
$\lambda_5 \lambda_{HS_2}^{(A)}$	2	2	2	2	2
$\lambda_5 T$	2	2	2	2	2
$\lambda_5 \lambda_{HS_1}^{(A)}$	-2	-2	-2	-2	-2
$\lambda_5 \lambda_{12}^{(A)}$	$-\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$
$\lambda_5 \lambda_{12}^{(\times)}$	0	0	0	0	0

Run block 3: T3-C-T3-C

$\mathcal{S}$	T3 - C ($\alpha = -2$)	T3 - C ($\alpha = -1$)	T3 - C ($\alpha = 0$)	T3 - C ($\alpha = 1$)	T3 - C ($\alpha = 2$)
$\lambda_5 \lambda_1$	2	2	2	2	2
$\lambda_5 \lambda_{HS_1}^{(1)}$	4	4	4	4	4
$\lambda_5 \lambda_{HS_2}^{(1)}$	4	4	4	4	4
$\lambda_5 \lambda_{12}^{(1)}$	2	2	2	2	2
$g_2^2 \lambda_5$	-9	-9	-9	-9	-9
$g_1^2 \lambda_5$	$-\frac{9}{2}$	-3	$-\frac{9}{2}$	-9	$-\frac{33}{2}$
$\lambda_5 \lambda_{HS_2}^{(A)}$	2	2	2	2	2
$\lambda_5 T$	2	2	2	2	2
$\lambda_5 \lambda_{HS_1}^{(A)}$	-2	-2	-2	-2	-2
$\lambda_5 \lambda_{12}^{(A)}$	$-\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$
$\lambda_5 \lambda_{12}^{(\times)}$	0	0	0	0	0

Run block 4: T3-D-T3-D

$\mathcal{S}$	T3 - D ($\alpha = -2$)	T3 - D ($\alpha = -1$)	T3 - D ($\alpha = 0$)	T3 - D ($\alpha = 1$)	T3 - D ($\alpha = 2$)
---------------	--------------------------	--------------------------	-------------------------	-------------------------	-------------------------

$\lambda_5 \lambda_1$	2	2	2	2	2
$\lambda_5 \lambda_{HS_1}^{(1)}$	4	4	4	4	4
$\lambda_5 \lambda_{HS_2}^{(1)}$	4	4	4	4	4
$\lambda_5 \lambda_{12}^{(1)}$	2	2	2	2	2
$g_2^2 \lambda_5$	$-\frac{21}{2}$	$-\frac{21}{2}$	$-\frac{21}{2}$	$-\frac{21}{2}$	$-\frac{21}{2}$
$g_1^2 \lambda_5$	$-\frac{9}{2}$	-3	$-\frac{9}{2}$	-9	$-\frac{33}{2}$
$\lambda_5 \lambda_{HS_2}^{(A)}$	0	0	0	0	0
$\lambda_5 T$	2	2	2	2	2
$\lambda_5 \lambda_{HS_1}^{(A)}$	-4	-4	-4	-4	-4
$\lambda_5 \lambda_{12}^{(A)}$	0	0	0	0	0
$\lambda_5 \lambda_{12}^{(\times)}$	0	0	0	0	0

Run block 5: T3-E-T3-E

$\mathcal{S}$	T3 - E ($\alpha = -2$)	T3 - E ($\alpha = -1$)	T3 - E ($\alpha = 0$)	T3 - E ($\alpha = 1$)	T3 - E ($\alpha = 2$)
$\lambda_5 \lambda_1$	2	2	2	2	2
$\lambda_5 \lambda_{HS_1}^{(1)}$	4	4	4	4	4
$\lambda_5 \lambda_{HS_2}^{(1)}$	4	4	4	4	4
$\lambda_5 \lambda_{12}^{(1)}$	2	2	2	2	2
$g_2^2 \lambda_5$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$
$g_1^2 \lambda_5$	$-\frac{9}{2}$	-3	$-\frac{9}{2}$	-9	$-\frac{33}{2}$
$\lambda_5 \lambda_{HS_2}^{(A)}$	2	2	2	2	2
$\lambda_5 T$	2	2	2	2	2
$\lambda_5 \lambda_{HS_1}^{(A)}$	-2	-2	-2	-2	-2
$\lambda_5 \lambda_{12}^{(A)}$	2	2	2	2	2
$\lambda_5 \lambda_{12}^{(\times)}$	-2	-2	-2	-2	-2

Additional scalar-coupling beta functions

$\beta_{\lambda_{HS_1}^{(A)}}$

Run block 1: T3-A-T3-A

$\mathcal{S}$	T3 - A ($\alpha = -2$)	T3 - A ($\alpha = -1$)	T3 - A ($\alpha = 0$)	T3 - A ($\alpha = 1$)	T3 - A ($\alpha = 2$)
$g_1^2 g_2^2$	0	0	0	0	0

Run block 2: T3-B-T3-B

$\mathcal{S}$	T3 - B ($\alpha = -2$)	T3 - B ($\alpha = -1$)	T3 - B ($\alpha = 0$)	T3 - B ($\alpha = 1$)	T3 - B ($\alpha = 2$)
$g_1^2 g_2^2$	-12	-6	0	6	12

Run block 3: T3-C-T3-C

$\mathcal{S}$	T3 - C ($\alpha = -2$)	T3 - C ($\alpha = -1$)	T3 - C ($\alpha = 0$)	T3 - C ($\alpha = 1$)	T3 - C ($\alpha = 2$)
$g_1^2 g_2^2$	-12	-6	0	6	12

Run block 4: T3-D-T3-D

$\mathcal{S}$	T3 - D ($\alpha = -2$)	T3 - D ($\alpha = -1$)	T3 - D ($\alpha = 0$)	T3 - D ($\alpha = 1$)	T3 - D ($\alpha = 2$)
$g_1^2 g_2^2$	-12	-6	0	6	12

Run block 5: T3-E-T3-E

S	T3 – E ($\alpha = -2$)	T3 – E ($\alpha = -1$)	T3 – E ($\alpha = 0$)	T3 – E ($\alpha = 1$)	T3 – E ($\alpha = 2$)
$g_1^2 g_2^2$	-12	-6	0	6	12

$$\beta_{\lambda_{HS_2}^{(A)}}$$

Run block 1: T3-A-T3-A

S	T3 – A ($\alpha = -2$)	T3 – A ($\alpha = -1$)	T3 – A ($\alpha = 0$)	T3 – A ($\alpha = 1$)	T3 – A ($\alpha = 2$)
$g_1^2 g_2^2$	0	6	12	18	24

Run block 2: T3-B-T3-B

S	T3 – B ($\alpha = -2$)	T3 – B ($\alpha = -1$)	T3 – B ($\alpha = 0$)	T3 – B ($\alpha = 1$)	T3 – B ($\alpha = 2$)
$g_1^2 g_2^2$	0	6	12	18	24

Run block 3: T3-C-T3-C

S	T3 – C ($\alpha = -2$)	T3 – C ($\alpha = -1$)	T3 – C ($\alpha = 0$)	T3 – C ($\alpha = 1$)	T3 – C ($\alpha = 2$)
$g_1^2 g_2^2$	0	6	12	18	24

Run block 4: T3-D-T3-D

S	T3 – D ($\alpha = -2$)	T3 – D ($\alpha = -1$)	T3 – D ($\alpha = 0$)	T3 – D ($\alpha = 1$)	T3 – D ($\alpha = 2$)
$g_1^2 g_2^2$	0	0	0	0	0

Run block 5: T3-E-T3-E

S	T3 – E ($\alpha = -2$)	T3 – E ($\alpha = -1$)	T3 – E ($\alpha = 0$)	T3 – E ($\alpha = 1$)	T3 – E ($\alpha = 2$)
$g_1^2 g_2^2$	0	6	12	18	24

$$\beta_{\lambda_{12}^{(A)}}$$

Run block 1: T3-A-T3-A

S	T3 – A ($\alpha = -2$)	T3 – A ($\alpha = -1$)	T3 – A ($\alpha = 0$)	T3 – A ($\alpha = 1$)	T3 – A ($\alpha = 2$)
$g_1^2 g_2^2$	0	0	0	0	0

Run block 2: T3-B-T3-B

S	T3 – B ($\alpha = -2$)	T3 – B ($\alpha = -1$)	T3 – B ($\alpha = 0$)	T3 – B ($\alpha = 1$)	T3 – B ($\alpha = 2$)
$g_1^2 g_2^2$	0	-6	0	18	48

Run block 3: T3-C-T3-C

S	T3 – C ($\alpha = -2$)	T3 – C ($\alpha = -1$)	T3 – C ($\alpha = 0$)	T3 – C ($\alpha = 1$)	T3 – C ($\alpha = 2$)
$g_1^2 g_2^2$	0	-6	0	18	48

Run block 4: T3-D-T3-D

$\mathcal{S}$	T3 – D ($\alpha = -2$)	T3 – D ($\alpha = -1$)	T3 – D ($\alpha = 0$)	T3 – D ($\alpha = 1$)	T3 – D ($\alpha = 2$)
$g_1^2 g_2^2$	0	0	0	0	0

Run block 5: T3-E–T3-E

$\mathcal{S}$	T3 – E ($\alpha = -2$)	T3 – E ($\alpha = -1$)	T3 – E ($\alpha = 0$)	T3 – E ($\alpha = 1$)	T3 – E ($\alpha = 2$)
$g_1^2 g_2^2$	0	−6	0	18	48

$\beta_{\lambda_{12}^{(\times)}}$

Run block 1: T3-A–T3-A

$\mathcal{S}$	T3 – A ($\alpha = -2$)	T3 – A ($\alpha = -1$)	T3 – A ($\alpha = 0$)	T3 – A ($\alpha = 1$)	T3 – A ($\alpha = 2$)
g_2^4	0	0	0	0	0
$\left(\lambda_{12}^{(A)}\right)^2$	0	0	0	0	0
$\lambda_{12}^{(1)} \lambda_{12}^{(\times)}$	0	0	0	0	0
$\left(\lambda_{12}^{(\times)}\right)^2$	0	0	0	0	0
$\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$	0	0	0	0	0
$\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$	0	0	0	0	0

Run block 2: T3-B–T3-B

$\mathcal{S}$	T3 – B ($\alpha = -2$)	T3 – B ($\alpha = -1$)	T3 – B ($\alpha = 0$)	T3 – B ($\alpha = 1$)	T3 – B ($\alpha = 2$)
g_2^4	0	0	0	0	0
$\left(\lambda_{12}^{(A)}\right)^2$	0	0	0	0	0
$\lambda_{12}^{(1)} \lambda_{12}^{(\times)}$	0	0	0	0	0
$\left(\lambda_{12}^{(\times)}\right)^2$	0	0	0	0	0
$\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$	0	0	0	0	0
$\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$	0	0	0	0	0

Run block 3: T3-C–T3-C

$\mathcal{S}$	T3 – C ($\alpha = -2$)	T3 – C ($\alpha = -1$)	T3 – C ($\alpha = 0$)	T3 – C ($\alpha = 1$)	T3 – C ($\alpha = 2$)
g_2^4	0	0	0	0	0
$\left(\lambda_{12}^{(A)}\right)^2$	0	0	0	0	0
$\lambda_{12}^{(1)} \lambda_{12}^{(\times)}$	0	0	0	0	0
$\left(\lambda_{12}^{(\times)}\right)^2$	0	0	0	0	0
$\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$	0	0	0	0	0
$\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$	0	0	0	0	0

Run block 4: T3-D–T3-D

$\mathcal{S}$	T3 – D ($\alpha = -2$)	T3 – D ($\alpha = -1$)	T3 – D ($\alpha = 0$)	T3 – D ($\alpha = 1$)	T3 – D ($\alpha = 2$)
g_2^4	0	0	0	0	0
$\left(\lambda_{12}^{(A)}\right)^2$	0	0	0	0	0
$\lambda_{12}^{(1)} \lambda_{12}^{(\times)}$	0	0	0	0	0
$\left(\lambda_{12}^{(\times)}\right)^2$	0	0	0	0	0

$\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$	0	0	0	0	0
$\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$	0	0	0	0	0

Run block 5: T3-E-T3-E

$\mathcal{S}$	T3 – E ($\alpha = -2$)	T3 – E ($\alpha = -1$)	T3 – E ($\alpha = 0$)	T3 – E ($\alpha = 1$)	T3 – E ($\alpha = 2$)
g_2^4	6	6	6	6	6
$\left(\lambda_{12}^{(A)}\right)^2$	2	2	2	2	2
$\lambda_{12}^{(1)} \lambda_{12}^{(\times)}$	8	8	8	8	8
$\left(\lambda_{12}^{(\times)}\right)^2$	10	10	10	10	10
$\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$	-2	-2	-2	-2	-2
$\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$	-2	-2	-2	-2	-2

Direct Weinberg mixing

The active EFT1 scalar interaction produces

$$16\pi^2 \beta_{C_5} \supset R_W \lambda_5 C_{LLS_1S_2}.$$

$\mathcal{S}$	T3 – A ($\alpha = -2$)	T3 – A ($\alpha = -1$)	T3 – A ($\alpha = 0$)	T3 – A ($\alpha = 1$)	T3 – A ($\alpha = 2$)
R_W	$\frac{4\sqrt{3}}{3}$	$\frac{4\sqrt{3}}{3}$	$\frac{4\sqrt{3}}{3}$	$\frac{4\sqrt{3}}{3}$	$\frac{4\sqrt{3}}{3}$

$\mathcal{S}$	T3 – B ($\alpha = -2$)	T3 – B ($\alpha = -1$)	T3 – B ($\alpha = 0$)	T3 – B ($\alpha = 1$)	T3 – B ($\alpha = 2$)
R_W	$\frac{4\sqrt{3}}{3}$	$\frac{4\sqrt{3}}{3}$	$\frac{4\sqrt{3}}{3}$	$\frac{4\sqrt{3}}{3}$	$\frac{4\sqrt{3}}{3}$

$\mathcal{S}$	T3 – C ($\alpha = -2$)	T3 – C ($\alpha = -1$)	T3 – C ($\alpha = 0$)	T3 – C ($\alpha = 1$)	T3 – C ($\alpha = 2$)
R_W	$-\frac{4}{3}$	$-\frac{4}{3}$	$-\frac{4}{3}$	$-\frac{4}{3}$	$-\frac{4}{3}$

$\mathcal{S}$	T3 – D ($\alpha = -2$)	T3 – D ($\alpha = -1$)	T3 – D ($\alpha = 0$)	T3 – D ($\alpha = 1$)	T3 – D ($\alpha = 2$)
R_W	$\frac{4\sqrt{3}}{3}$	$\frac{4\sqrt{3}}{3}$	$\frac{4\sqrt{3}}{3}$	$\frac{4\sqrt{3}}{3}$	$\frac{4\sqrt{3}}{3}$

$\mathcal{S}$	T3 – E ($\alpha = -2$)	T3 – E ($\alpha = -1$)	T3 – E ($\alpha = 0$)	T3 – E ($\alpha = 1$)	T3 – E ($\alpha = 2$)
R_W	$-\frac{4\sqrt{6}}{3}$	$-\frac{4\sqrt{6}}{3}$	$-\frac{4\sqrt{6}}{3}$	$-\frac{4\sqrt{6}}{3}$	$-\frac{4\sqrt{6}}{3}$

The displayed R_W values use the validated Wigner-6j reduction and inherited tensor convention. The overall tensor-level normalization/phase has not yet been independently derived from first principles.

F-threshold matching tensor samples

The component bases have representation-dependent dimensions, so the raw tensor entries are not placed in the cross-run beta-function tables.

T3 – A ($\alpha = -2$)

48 non-zero real-basis tensor components reconstructed; showing the first 10.

C_{ijab}	matched coefficient
C_{1157}	$\frac{3^{\frac{3}{4}} \overline{y_1 y_2}}{12 M_F}$
C_{1158}	$\frac{3^{\frac{3}{4}} i \overline{y_1 y_2}}{12 M_F}$
C_{11511}	$\frac{3^{\frac{3}{4}} \overline{y_1 y_2}}{12 M_F}$
C_{11512}	$-\frac{3^{\frac{3}{4}} \overline{y_1 y_2}}{12 M_F}$
C_{1167}	$-\frac{3^{\frac{3}{4}} i \overline{y_1 y_2}}{12 M_F}$
C_{1168}	$\frac{3^{\frac{3}{4}} \overline{y_1 y_2}}{12 M_F}$
C_{11611}	$\frac{3^{\frac{3}{4}} \overline{y_1 y_2}}{12 M_F}$
C_{11612}	$\frac{3^{\frac{3}{4}} i \overline{y_1 y_2}}{12 M_F}$
C_{1175}	$\frac{3^{\frac{3}{4}} \overline{y_1 y_2}}{12 M_F}$
C_{1176}	$-\frac{3^{\frac{3}{4}} i \overline{y_1 y_2}}{12 M_F}$

T3 – A ($\alpha = -1$)

48 non-zero real-basis tensor components reconstructed; showing the first 10.

C_{ijab}	matched coefficient
C_{1157}	$\frac{3^{\frac{3}{4}} \overline{y_1 y_2}}{12 M_F}$
C_{1158}	$\frac{3^{\frac{3}{4}} i \overline{y_1 y_2}}{12 M_F}$
C_{11511}	$\frac{3^{\frac{3}{4}} \overline{y_1 y_2}}{12 M_F}$
C_{11512}	$-\frac{3^{\frac{3}{4}} \overline{y_1 y_2}}{12 M_F}$
C_{1167}	$-\frac{3^{\frac{3}{4}} i \overline{y_1 y_2}}{12 M_F}$
C_{1168}	$\frac{3^{\frac{3}{4}} \overline{y_1 y_2}}{12 M_F}$
C_{11611}	$\frac{3^{\frac{3}{4}} \overline{y_1 y_2}}{12 M_F}$
C_{11612}	$\frac{3^{\frac{3}{4}} i \overline{y_1 y_2}}{12 M_F}$
C_{1175}	$\frac{3^{\frac{3}{4}} \overline{y_1 y_2}}{12 M_F}$
C_{1176}	$-\frac{3^{\frac{3}{4}} i \overline{y_1 y_2}}{12 M_F}$

T3 – A ($\alpha = 0$)

48 non-zero real-basis tensor components reconstructed; showing the first 10.

C_{ijab}	matched coefficient
C_{1157}	$\frac{3^{\frac{3}{4}} \overline{y_1 y_2}}{12 M_F}$
C_{1158}	$\frac{3^{\frac{3}{4}} i \overline{y_1 y_2}}{12 M_F}$
C_{11511}	$\frac{3^{\frac{3}{4}} \overline{y_1 y_2}}{12 M_F}$
C_{11512}	$-\frac{3^{\frac{3}{4}} \overline{y_1 y_2}}{12 M_F}$
C_{1167}	$-\frac{3^{\frac{3}{4}} i \overline{y_1 y_2}}{12 M_F}$
C_{1168}	$\frac{3^{\frac{3}{4}} \overline{y_1 y_2}}{12 M_F}$
C_{11611}	$\frac{3^{\frac{3}{4}} \overline{y_1 y_2}}{12 M_F}$
C_{11612}	$\frac{3^{\frac{3}{4}} i \overline{y_1 y_2}}{12 M_F}$
C_{1175}	$\frac{3^{\frac{3}{4}} \overline{y_1 y_2}}{12 M_F}$
C_{1176}	$-\frac{3^{\frac{3}{4}} i \overline{y_1 y_2}}{12 M_F}$

T3 – A ($\alpha = 1$)

48 non-zero real-basis tensor components reconstructed; showing the first 10.

C_{ijab}	matched coefficient
C_{1157}	$\frac{3^{\frac{3}{4}} \overline{y_1 y_2}}{12 M_F}$
C_{1158}	$\frac{3^{\frac{3}{4}} i \overline{y_1 y_2}}{12 M_F}$
C_{11511}	$\frac{3^{\frac{3}{4}} i \overline{y_1 y_2}}{12 M_F}$
C_{11512}	$-\frac{3^{\frac{3}{4}} \overline{y_1 y_2}}{12 M_F}$
C_{1167}	$-\frac{3^{\frac{3}{4}} i \overline{y_1 y_2}}{12 M_F}$
C_{1168}	$\frac{3^{\frac{3}{4}} \overline{y_1 y_2}}{12 M_F}$
C_{11611}	$\frac{3^{\frac{3}{4}} \overline{y_1 y_2}}{12 M_F}$
C_{11612}	$\frac{3^{\frac{3}{4}} i \overline{y_1 y_2}}{12 M_F}$
C_{1175}	$\frac{3^{\frac{3}{4}} \overline{y_1 y_2}}{12 M_F}$
C_{1176}	$-\frac{3^{\frac{3}{4}} i \overline{y_1 y_2}}{12 M_F}$

T3 – A ($\alpha = 2$)

48 non-zero real-basis tensor components reconstructed; showing the first 10.

C_{ijab}	matched coefficient
C_{1157}	$\frac{3^{\frac{3}{4}} \overline{y_1 y_2}}{12 M_F}$
C_{1158}	$\frac{3^{\frac{3}{4}} i \overline{y_1 y_2}}{12 M_F}$
C_{11511}	$\frac{3^{\frac{3}{4}} i \overline{y_1 y_2}}{12 M_F}$
C_{11512}	$-\frac{3^{\frac{3}{4}} \overline{y_1 y_2}}{12 M_F}$
C_{1167}	$-\frac{3^{\frac{3}{4}} i \overline{y_1 y_2}}{12 M_F}$
C_{1168}	$\frac{3^{\frac{3}{4}} \overline{y_1 y_2}}{12 M_F}$
C_{11611}	$\frac{3^{\frac{3}{4}} \overline{y_1 y_2}}{12 M_F}$
C_{11612}	$\frac{3^{\frac{3}{4}} i \overline{y_1 y_2}}{12 M_F}$
C_{1175}	$\frac{3^{\frac{3}{4}} \overline{y_1 y_2}}{12 M_F}$
C_{1176}	$-\frac{3^{\frac{3}{4}} i \overline{y_1 y_2}}{12 M_F}$

T3 – B ($\alpha = -2$)

48 non-zero real-basis tensor components reconstructed; showing the first 10.

C_{ijab}	matched coefficient
C_{11511}	$\frac{\overline{y_1 y_2}}{4 M_F}$
C_{11512}	$\frac{i \overline{y_1 y_2}}{4 M_F}$
C_{11611}	$-\frac{i \overline{y_1 y_2}}{4 M_F}$
C_{11612}	$\frac{\overline{y_1 y_2}}{4 M_F}$
C_{11115}	$\frac{\overline{y_1 y_2}}{4 M_F}$
C_{11116}	$-\frac{i \overline{y_1 y_2}}{4 M_F}$
C_{11125}	$\frac{i \overline{y_1 y_2}}{4 M_F}$
C_{11126}	$\frac{\overline{y_1 y_2}}{4 M_F}$
C_{1259}	$-\frac{\overline{y_1 y_2}}{8 M_F}$
C_{12510}	$-\frac{i \overline{y_1 y_2}}{8 M_F}$

T3 – B ($\alpha = -1$)

96 non-zero real-basis tensor components reconstructed; showing the first 10.

C_{ijab}	matched coefficient
C_{1155}	$\frac{\overline{y_1}^2}{4M_F}$
C_{1156}	$-\frac{i\overline{y_1}}{4M_F}$
C_{11511}	$\frac{\overline{y_1}y_2}{4M_F}$
C_{11512}	$\frac{i\overline{y_1}y_2}{4M_F}$
C_{1165}	$-\frac{i\overline{y_1}}{4M_F}$
C_{1166}	$-\frac{\overline{y_1}^2}{4M_F}$
C_{11611}	$-\frac{i\overline{y_1}y_2}{4M_F}$
C_{11612}	$\frac{\overline{y_1}y_2}{4M_F}$
C_{11115}	$\frac{\overline{y_1}y_2}{4M_F}$
C_{11116}	$-\frac{i\overline{y_1}y_2}{4M_F}$

T3 – B ($\alpha = 0$)

48 non-zero real-basis tensor components reconstructed; showing the first 10.

C_{ijab}	matched coefficient
C_{11511}	$\frac{\overline{y_1}y_2}{4M_F}$
C_{11512}	$\frac{i\overline{y_1}y_2}{4M_F}$
C_{11611}	$-\frac{i\overline{y_1}y_2}{4M_F}$
C_{11612}	$\frac{\overline{y_1}y_2}{4M_F}$
C_{11115}	$\frac{\overline{y_1}y_2}{4M_F}$
C_{11116}	$-\frac{i\overline{y_1}y_2}{4M_F}$
C_{11125}	$\frac{i\overline{y_1}y_2}{4M_F}$
C_{11126}	$\frac{\overline{y_1}y_2}{4M_F}$
C_{1259}	$-\frac{\overline{y_1}y_2}{8M_F}$
C_{12510}	$-\frac{i\overline{y_1}y_2}{8M_F}$

T3 – B ($\alpha = 1$)

48 non-zero real-basis tensor components reconstructed; showing the first 10.

C_{ijab}	matched coefficient
C_{11511}	$\frac{\overline{y_1}y_2}{4M_F}$
C_{11512}	$\frac{i\overline{y_1}y_2}{4M_F}$
C_{11611}	$-\frac{i\overline{y_1}y_2}{4M_F}$
C_{11612}	$\frac{\overline{y_1}y_2}{4M_F}$
C_{11115}	$\frac{\overline{y_1}y_2}{4M_F}$
C_{11116}	$-\frac{i\overline{y_1}y_2}{4M_F}$
C_{11125}	$\frac{i\overline{y_1}y_2}{4M_F}$
C_{11126}	$\frac{\overline{y_1}y_2}{4M_F}$
C_{1259}	$-\frac{\overline{y_1}y_2}{8M_F}$
C_{12510}	$-\frac{i\overline{y_1}y_2}{8M_F}$

T3 – B ($\alpha = 2$)

48 non-zero real-basis tensor components reconstructed; showing the first 10.

C_{ijab}	matched coefficient
------------	---------------------

C_{11511}	$\frac{\overline{y_1 y_2}}{4M_F}$
C_{11512}	$\frac{i\overline{y_1 y_2}}{4M_F}$
C_{11611}	$-\frac{i\overline{y_1 y_2}}{4M_F}$
C_{11612}	$\frac{\overline{y_1 y_2}}{4M_F}$
C_{11115}	$\frac{\overline{y_1 y_2}}{4M_F}$
C_{11116}	$-\frac{i\overline{y_1 y_2}}{4M_F}$
C_{11125}	$\frac{i\overline{y_1 y_2}}{4M_F}$
C_{11126}	$\frac{\overline{y_1 y_2}}{4M_F}$
C_{1259}	$-\frac{\overline{y_1 y_2}}{8M_F}$
C_{12510}	$-\frac{i\overline{y_1 y_2}}{8M_F}$

T3 - C ($\alpha = -2$)

48 non-zero real-basis tensor components reconstructed; showing the first 10.

C_{ijab}	matched coefficient
C_{11511}	$-\frac{\sqrt{3}\overline{y_1 y_2}}{12M_F}$
C_{11512}	$-\frac{\sqrt{3}i\overline{y_1 y_2}}{12M_F}$
C_{11611}	$\frac{\sqrt{3}i\overline{y_1 y_2}}{12M_F}$
C_{11612}	$-\frac{\sqrt{3}\overline{y_1 y_2}}{12M_F}$
C_{11115}	$-\frac{\sqrt{3}\overline{y_1 y_2}}{12M_F}$
C_{11116}	$\frac{\sqrt{3}i\overline{y_1 y_2}}{12M_F}$
C_{11125}	$-\frac{\sqrt{3}i\overline{y_1 y_2}}{12M_F}$
C_{11126}	$-\frac{\sqrt{3}\overline{y_1 y_2}}{12M_F}$
C_{1259}	$\frac{\sqrt{3}\overline{y_1 y_2}}{24M_F}$
C_{12510}	$\frac{\sqrt{3}i\overline{y_1 y_2}}{24M_F}$

T3 - C ($\alpha = -1$)

96 non-zero real-basis tensor components reconstructed; showing the first 10.

C_{ijab}	matched coefficient
C_{1155}	$\frac{\sqrt{3}\overline{y_1}^2}{12M_F}$
C_{1156}	$-\frac{\sqrt{3}i\overline{y_1}^2}{12M_F}$
C_{11511}	$-\frac{\sqrt{3}\overline{y_1 y_2}}{12M_F}$
C_{11512}	$-\frac{\sqrt{3}i\overline{y_1 y_2}}{12M_F}$
C_{1165}	$-\frac{\sqrt{3}i\overline{y_1}^2}{12M_F}$
C_{1166}	$-\frac{\sqrt{3}\overline{y_1}^2}{12M_F}$
C_{11611}	$\frac{\sqrt{3}i\overline{y_1 y_2}}{12M_F}$
C_{11612}	$-\frac{\sqrt{3}\overline{y_1 y_2}}{12M_F}$
C_{11115}	$-\frac{\sqrt{3}\overline{y_1 y_2}}{12M_F}$
C_{11116}	$\frac{\sqrt{3}i\overline{y_1 y_2}}{12M_F}$

T3 - C ($\alpha = 0$)

48 non-zero real-basis tensor components reconstructed; showing the first 10.

C_{ijab}	matched coefficient
C_{11511}	$-\frac{\sqrt{3}\overline{y_1 y_2}}{12M_F}$
C_{11512}	$-\frac{\sqrt{3}i\overline{y_1 y_2}}{12M_F}$
C_{11611}	$\frac{\sqrt{3}i\overline{y_1 y_2}}{12M_F}$

C_{11612}	$-\frac{\sqrt{3}y_1y_2}{12M_F}$
C_{11115}	$-\frac{\sqrt{3}y_1y_2}{12M_F}$
C_{11116}	$\frac{\sqrt{3}iy_1y_2}{12M_F}$
C_{11125}	$-\frac{\sqrt{3}iy_1y_2}{12M_F}$
C_{11126}	$-\frac{\sqrt{3}y_1y_2}{12M_F}$
C_{1259}	$\frac{\sqrt{3}y_1y_2}{24M_F}$
C_{12510}	$\frac{\sqrt{3}iy_1y_2}{24M_F}$

T3 – C ($\alpha = 1$)

48 non-zero real-basis tensor components reconstructed; showing the first 10.

C_{ijab}	matched coefficient
C_{11511}	$-\frac{\sqrt{3}y_1y_2}{12M_F}$
C_{11512}	$-\frac{\sqrt{3}iy_1y_2}{12M_F}$
C_{11611}	$\frac{\sqrt{3}iy_1y_2}{12M_F}$
C_{11612}	$-\frac{\sqrt{3}y_1y_2}{12M_F}$
C_{11115}	$-\frac{\sqrt{3}y_1y_2}{12M_F}$
C_{11116}	$\frac{\sqrt{3}iy_1y_2}{12M_F}$
C_{11125}	$-\frac{\sqrt{3}iy_1y_2}{12M_F}$
C_{11126}	$-\frac{\sqrt{3}y_1y_2}{12M_F}$
C_{1259}	$\frac{\sqrt{3}y_1y_2}{24M_F}$
C_{12510}	$\frac{\sqrt{3}iy_1y_2}{24M_F}$

T3 – C ($\alpha = 2$)

48 non-zero real-basis tensor components reconstructed; showing the first 10.

C_{ijab}	matched coefficient
C_{11511}	$-\frac{\sqrt{3}y_1y_2}{12M_F}$
C_{11512}	$-\frac{\sqrt{3}iy_1y_2}{12M_F}$
C_{11611}	$\frac{\sqrt{3}iy_1y_2}{12M_F}$
C_{11612}	$-\frac{\sqrt{3}y_1y_2}{12M_F}$
C_{11115}	$-\frac{\sqrt{3}y_1y_2}{12M_F}$
C_{11116}	$\frac{\sqrt{3}iy_1y_2}{12M_F}$
C_{11125}	$-\frac{\sqrt{3}iy_1y_2}{12M_F}$
C_{11126}	$-\frac{\sqrt{3}y_1y_2}{12M_F}$
C_{1259}	$\frac{\sqrt{3}y_1y_2}{24M_F}$
C_{12510}	$\frac{\sqrt{3}iy_1y_2}{24M_F}$

T3 – D ($\alpha = -2$)

48 non-zero real-basis tensor components reconstructed; showing the first 10.

C_{ijab}	matched coefficient
C_{11511}	$\frac{3^{\frac{3}{4}}y_1y_2}{12M_F}$
C_{11512}	$\frac{3^{\frac{3}{4}}iy_1y_2}{12M_F}$
C_{11611}	$-\frac{3^{\frac{3}{4}}iy_1y_2}{12M_F}$
C_{11612}	$\frac{3^{\frac{3}{4}}y_1y_2}{12M_F}$
C_{11911}	$\frac{3^{\frac{3}{4}}iy_1y_2}{12M_F}$

C_{11912}	$-\frac{3^{\frac{3}{4}}\overline{y_1 y_2}}{12M_F}$
C_{111011}	$\frac{3^{\frac{3}{4}}\overline{y_1 y_2}}{12M_F}$
C_{111012}	$\frac{3^{\frac{3}{4}}i\overline{y_1 y_2}}{12M_F}$
C_{11115}	$\frac{3^{\frac{3}{4}}\overline{y_1 y_2}}{12M_F}$
C_{11116}	$-\frac{3^{\frac{3}{4}}i\overline{y_1 y_2}}{12M_F}$

T3 – D ($\alpha = -1$)

48 non-zero real-basis tensor components reconstructed; showing the first 10.

C_{ijab}	matched coefficient
C_{11511}	$\frac{3^{\frac{3}{4}}\overline{y_1 y_2}}{12M_F}$
C_{11512}	$\frac{3^{\frac{3}{4}}i\overline{y_1 y_2}}{12M_F}$
C_{11611}	$-\frac{3^{\frac{3}{4}}i\overline{y_1 y_2}}{12M_F}$
C_{11612}	$\frac{3^{\frac{3}{4}}\overline{y_1 y_2}}{12M_F}$
C_{11911}	$\frac{3^{\frac{3}{4}}i\overline{y_1 y_2}}{12M_F}$
C_{11912}	$-\frac{3^{\frac{3}{4}}\overline{y_1 y_2}}{12M_F}$
C_{111011}	$\frac{3^{\frac{3}{4}}\overline{y_1 y_2}}{12M_F}$
C_{111012}	$\frac{3^{\frac{3}{4}}i\overline{y_1 y_2}}{12M_F}$
C_{11115}	$\frac{3^{\frac{3}{4}}\overline{y_1 y_2}}{12M_F}$
C_{11116}	$-\frac{3^{\frac{3}{4}}i\overline{y_1 y_2}}{12M_F}$

T3 – D ($\alpha = 0$)

48 non-zero real-basis tensor components reconstructed; showing the first 10.

C_{ijab}	matched coefficient
C_{11511}	$\frac{3^{\frac{3}{4}}\overline{y_1 y_2}}{12M_F}$
C_{11512}	$\frac{3^{\frac{3}{4}}i\overline{y_1 y_2}}{12M_F}$
C_{11611}	$-\frac{3^{\frac{3}{4}}i\overline{y_1 y_2}}{12M_F}$
C_{11612}	$\frac{3^{\frac{3}{4}}\overline{y_1 y_2}}{12M_F}$
C_{11911}	$\frac{3^{\frac{3}{4}}i\overline{y_1 y_2}}{12M_F}$
C_{11912}	$-\frac{3^{\frac{3}{4}}\overline{y_1 y_2}}{12M_F}$
C_{111011}	$\frac{3^{\frac{3}{4}}\overline{y_1 y_2}}{12M_F}$
C_{111012}	$\frac{3^{\frac{3}{4}}i\overline{y_1 y_2}}{12M_F}$
C_{11115}	$\frac{3^{\frac{3}{4}}\overline{y_1 y_2}}{12M_F}$
C_{11116}	$-\frac{3^{\frac{3}{4}}i\overline{y_1 y_2}}{12M_F}$

T3 – D ($\alpha = 1$)

48 non-zero real-basis tensor components reconstructed; showing the first 10.

C_{ijab}	matched coefficient
C_{11511}	$\frac{3^{\frac{3}{4}}\overline{y_1 y_2}}{12M_F}$
C_{11512}	$\frac{3^{\frac{3}{4}}i\overline{y_1 y_2}}{12M_F}$
C_{11611}	$-\frac{3^{\frac{3}{4}}i\overline{y_1 y_2}}{12M_F}$

C_{11612}	$\frac{3^{\frac{3}{4}} \overline{y_1 y_2}}{12 M_F}$
C_{11911}	$\frac{3^{\frac{3}{4}} i \overline{y_1 y_2}}{12 M_F}$
C_{11912}	$-\frac{3^{\frac{3}{4}} \overline{y_1 y_2}}{12 M_F}$
C_{111011}	$\frac{3^{\frac{3}{4}} \overline{y_1 y_2}}{12 M_F}$
C_{111012}	$\frac{3^{\frac{3}{4}} i \overline{y_1 y_2}}{12 M_F}$
C_{11115}	$\frac{3^{\frac{3}{4}} \overline{y_1 y_2}}{12 M_F}$
C_{11116}	$-\frac{3^{\frac{3}{4}} i \overline{y_1 y_2}}{12 M_F}$

T3 – D ($\alpha = 2$)

48 non-zero real-basis tensor components reconstructed; showing the first 10.

C_{ijab}	matched coefficient
C_{11511}	$\frac{3^{\frac{3}{4}} \overline{y_1 y_2}}{12 M_F}$
C_{11512}	$\frac{3^{\frac{3}{4}} i \overline{y_1 y_2}}{12 M_F}$
C_{11611}	$-\frac{3^{\frac{3}{4}} i \overline{y_1 y_2}}{12 M_F}$
C_{11612}	$\frac{3^{\frac{3}{4}} \overline{y_1 y_2}}{12 M_F}$
C_{11911}	$\frac{3^{\frac{3}{4}} i \overline{y_1 y_2}}{12 M_F}$
C_{11912}	$-\frac{3^{\frac{3}{4}} \overline{y_1 y_2}}{12 M_F}$
C_{111011}	$\frac{3^{\frac{3}{4}} \overline{y_1 y_2}}{12 M_F}$
C_{111012}	$\frac{3^{\frac{3}{4}} i \overline{y_1 y_2}}{12 M_F}$
C_{11115}	$\frac{3^{\frac{3}{4}} \overline{y_1 y_2}}{12 M_F}$
C_{11116}	$-\frac{3^{\frac{3}{4}} i \overline{y_1 y_2}}{12 M_F}$

T3 – E ($\alpha = -2$)

96 non-zero real-basis tensor components reconstructed; showing the first 10.

C_{ijab}	matched coefficient
C_{11513}	$\frac{\sqrt{3} \overline{y_1 y_2}}{12 M_F}$
C_{11514}	$\frac{\sqrt{3} i \overline{y_1 y_2}}{12 M_F}$
C_{11613}	$-\frac{\sqrt{3} i \overline{y_1 y_2}}{12 M_F}$
C_{11614}	$\frac{\sqrt{3} \overline{y_1 y_2}}{12 M_F}$
C_{11711}	$\frac{\sqrt{3} \overline{y_1 y_2}}{12 M_F}$
C_{11712}	$-\frac{\sqrt{3} i \overline{y_1 y_2}}{12 M_F}$
C_{11715}	$-\frac{\sqrt{3} i \overline{y_1 y_2}}{12 M_F}$
C_{11716}	$\frac{\sqrt{3} \overline{y_1 y_2}}{12 M_F}$
C_{11811}	$\frac{\sqrt{3} i \overline{y_1 y_2}}{12 M_F}$
C_{11812}	$-\frac{\sqrt{3} \overline{y_1 y_2}}{12 M_F}$

T3 – E ($\alpha = -1$)

96 non-zero real-basis tensor components reconstructed; showing the first 10.

C_{ijab}	matched coefficient
C_{11513}	$\frac{\sqrt{3} \overline{y_1 y_2}}{12 M_F}$
C_{11514}	$\frac{\sqrt{3} i \overline{y_1 y_2}}{12 M_F}$
C_{11613}	$-\frac{\sqrt{3} i \overline{y_1 y_2}}{12 M_F}$

C_{11614}	$\frac{\sqrt{3}y_1y_2}{12M_F}$
C_{11711}	$-\frac{\sqrt{3}y_1y_2}{12M_F}$
C_{11712}	$-\frac{\sqrt{3iy_1y_2}}{12M_F}$
C_{11715}	$-\frac{\sqrt{3iy_1y_2}}{12M_F}$
C_{11716}	$\frac{\sqrt{3}y_1y_2}{12M_F}$
C_{11811}	$\frac{\sqrt{3iy_1y_2}}{12M_F}$
C_{11812}	$-\frac{\sqrt{3}y_1y_2}{12M_F}$

T3 – E ($\alpha = 0$)

96 non-zero real-basis tensor components reconstructed; showing the first 10.

C_{ijab}	matched coefficient
C_{11513}	$\frac{\sqrt{3}y_1y_2}{12M_F}$
C_{11514}	$\frac{\sqrt{3iy_1y_2}}{12M_F}$
C_{11613}	$-\frac{\sqrt{3iy_1y_2}}{12M_F}$
C_{11614}	$\frac{\sqrt{3}y_1y_2}{12M_F}$
C_{11711}	$-\frac{\sqrt{3}y_1y_2}{12M_F}$
C_{11712}	$-\frac{\sqrt{3iy_1y_2}}{12M_F}$
C_{11715}	$-\frac{\sqrt{3iy_1y_2}}{12M_F}$
C_{11716}	$\frac{\sqrt{3}y_1y_2}{12M_F}$
C_{11811}	$\frac{\sqrt{3iy_1y_2}}{12M_F}$
C_{11812}	$-\frac{\sqrt{3}y_1y_2}{12M_F}$

T3 – E ($\alpha = 1$)

96 non-zero real-basis tensor components reconstructed; showing the first 10.

C_{ijab}	matched coefficient
C_{11513}	$\frac{\sqrt{3}y_1y_2}{12M_F}$
C_{11514}	$\frac{\sqrt{3iy_1y_2}}{12M_F}$
C_{11613}	$-\frac{\sqrt{3iy_1y_2}}{12M_F}$
C_{11614}	$\frac{\sqrt{3}y_1y_2}{12M_F}$
C_{11711}	$-\frac{\sqrt{3}y_1y_2}{12M_F}$
C_{11712}	$-\frac{\sqrt{3iy_1y_2}}{12M_F}$
C_{11715}	$-\frac{\sqrt{3iy_1y_2}}{12M_F}$
C_{11716}	$\frac{\sqrt{3}y_1y_2}{12M_F}$
C_{11811}	$\frac{\sqrt{3iy_1y_2}}{12M_F}$
C_{11812}	$-\frac{\sqrt{3}y_1y_2}{12M_F}$

T3 – E ($\alpha = 2$)

96 non-zero real-basis tensor components reconstructed; showing the first 10.

C_{ijab}	matched coefficient
C_{11513}	$\frac{\sqrt{3}y_1y_2}{12M_F}$
C_{11514}	$\frac{\sqrt{3iy_1y_2}}{12M_F}$
C_{11613}	$-\frac{\sqrt{3iy_1y_2}}{12M_F}$
C_{11614}	$\frac{\sqrt{3}y_1y_2}{12M_F}$
C_{11711}	$-\frac{\sqrt{3}y_1y_2}{12M_F}$

$$\begin{array}{l}
C_{11712} \quad -\frac{\sqrt{3}iy_1y_2}{12M_F} \\
C_{11715} \quad -\frac{\sqrt{3}iy_1y_2}{12M_F} \\
C_{11716} \quad \frac{\sqrt{3}y_1y_2}{12M_F} \\
C_{11811} \quad \frac{\sqrt{3}iy_1y_2}{12M_F} \\
C_{11812} \quad -\frac{\sqrt{3}y_1y_2}{12M_F}
\end{array}

---$$

Next threshold

Further matching depends on the next field or fields selected by the threshold plan. Once all T3 fields are removed, their representation dependence is carried by the matched C_5 boundary condition and the subsequent SMEFT running is universal.